

Pablo J. Diego Simón

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Education

Oct 2023 – Now

Ph.D. Deep Learning and Computational Neuroscience

AI4theSciences, Marie Skłodowska-Curie Actions (MSCA), Scholarship

Département d'Études Cognitives (DEC)

École Normale Supérieure (ENS), Paris

Thesis: *Linking Brain Dynamics and Large Language Models*

Sep 2021 – Sep 2023

M.Sc. Biomedical Engineering

Information Technology Department (D-ITET)

ETH Zürich

Courses and scores

Thesis: *Trainability and Memory Capacity in Recurrent Neural Networks.*

Sep 2017 – July 2021

B.Sc. Biomedical Engineering

Information Technology Department (ETSIT)

Universidad Rey Juan Carlos, Madrid

Courses and scores

Thesis: *A Bayesian Machine Scientist for Complex Systems' Symbolic Regression.*

Research Publications

Conference Proceedings

- 1 Pablo J. Diego-Simón and E. Niklasson, "Learning local universes: Neural Cellular Automata as decentralized world models," in *Paradigms of Intelligence, Google Research*, 2026.
- 2 P. Orhan, Pablo J. Diego-Simón, E. Chemla, Y. Boubenec, Y. Lakretz, and J.-R. King, "The Emergence of Phonemic, Syntactic, and Semantic Structures in Neural Networks: A Unified Geometrical Framework," in *Under revision.*, 2025.
- 3 Pablo J. Diego-Simón, E. Chemla, J.-R. King, and Y. Lakretz, "Probing Syntax in Large Language Models: Successes and Remaining Challenges," in *2nd Conference of Language Modeling (COLM 2025)*, 2025.
- 4 Pablo J. Diego-Simón, P. Orhan, E. Chemla, Y. Lakretz, and J.-R. King, "Polar probe decodes semantic structures from LLMs," in *2nd Conference of Language Modeling (COLM 2026)*, 2025.
- 5 Pablo J. Diego-Simón, S. D'Ascoli, E. Chemla, Y. Lakretz, and J.-R. King, "A Polar Coordinate System Represents Syntax in Language Models," in *38th Conference on Neural Information Processing Systems (NeurIPS 2024)*, 2024.
- 6 R. J. Algayres, Pablo J. Diego Simon, B. Sagot, and E. Dupoux, "XLS-r fine-tuning on noisy word boundaries for unsupervised speech segmentation into words," in *The 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. [URL: https://openreview.net/forum?id=FGBWdf7Z19](https://openreview.net/forum?id=FGBWdf7Z19).

Teaching

2026 (Lecturer) **Advanced Data Analysis**, ENS Ulm, Paris

Work experience

May 2026 – Sep 2026 **Research Scientist Intern**
Paradigms of Intelligence
Google Research, Zürich

Sep 2022 – Feb 2023 **Research Intern**
Cognitive Machine Learning lab
École Normale Supérieure (ENS), Paris

Jan 2022 – June 2022 **Research collaborator**
Google Research, Zürich

Awards and Certifications

Awards

2026 **Selected Participant, Declaration of Digital Duties and Responsibilities**, Fundación Hermes

2023 **Rafael Del Pino Scholarship, From Science to Society Workshop**, Christ College, Oxford

2021 **Award for the Outstanding Graduate of the Year**, Universidad Rey Juan Carlos, Madrid.
Award for the Best B.Sc. Thesis of the Year, Universidad Rey Juan Carlos, Madrid.

2020 **MIT Leading Digital Transformation Scholarship (Banco Santander)**, MIT PE.
Excellence Scholarship, Community of Madrid, Education Delegation.

2019 **Excellence Scholarship**, Community of Madrid, Education Delegation.
Honour Delegate Decoration, CMU Barberán y Collar.

2017 **10k Fundraising for Mekong Plus Association**, Institut Saint-André Ixelles.

Certifications

2020 **IELTS**. 8.0 Score.

2019 **Test de Français International (TFI)**. Maximum Score.

Skills

Coding	Python, Matlab, Bash, Javascript, SQL, C ...
Python Libraries	PyTorch, TensorFlow, Jax, Pandas, Networkx, Numpy ...
Languages	Spanish - Native English - Proficient French - Proficient German - Basic